AGROINTEL AI — Research Plan

Global demonstration

Vegetable agriculture

Project title: AGROINTEL AI: An Explainable AI-Powered Crop Rotation and Climate Resilience Decision-Support Platform

Research focus: Using NASA Earth observation data, agricultural science and intelligent decision-support techniques to identify vegetable crop-rotation strategies that can improve soil health, conserve water and strengthen resilience to climate variability.

1. Research overview

1.1 Background

Agriculture is increasingly affected by changing rainfall patterns, rising temperatures, water scarcity, soil degradation and extreme weather events.

Vegetable farming is particularly challenging because different vegetable crops have different growing periods, nutrient requirements, water requirements and susceptibility to pests and diseases.

Traditional crop-rotation decisions often depend on farmers' experience and local knowledge. Although this knowledge is valuable, combining it with satellite observations and environmental datasets could provide additional evidence for making better decisions.

AGROINTEL AI will investigate how NASA Earth observations and agricultural information can be combined into an explainable decision-support system that evaluates alternative vegetable crop-rotation strategies.

1.2 Research problem

The central research problem is:

How can satellite-derived environmental information and agricultural knowledge be used to identify vegetable crop-rotation strategies that improve soil-health potential and climate resilience while considering farmers' practical needs?

The project addresses five specific problems:

1. Limited integration of environmental satellite data into everyday crop-rotation decisions.
2. Difficulty identifying periods of increased heat, rainfall and water stress.

3. Lack of accessible comparisons between alternative crop rotations.
4. Limited consideration of soil improvement, pest-cycle disruption and water requirements together.
5. Difficulty communicating complex environmental information to farmers.

2. Research objectives

The project will have six measurable objectives.

No.	Objective
1	Collect and analyse NASA environmental datasets.
2	Identify historical climate and vegetation patterns relevant to vegetable farming.
3	Develop an explainable crop-rotation scoring model.
4	Compare at least three vegetable crop-rotation strategies.
5	Develop an interactive global agricultural intelligence platform.
6	Evaluate the system's usefulness, scientific reliability and limitations.

3. Research questions and hypotheses

The research will investigate the following questions:

Research question	Investigation
RQ1	Can NASA observations identify environmental conditions that increase vegetable crop vulnerability?
RQ2	How do rainfall, temperature and soil moisture patterns influence rotation suitability?
RQ3	Can legume integration improve the soil-health potential of vegetable rotations?
RQ4	Which rotation strategy offers the best balance between water conservation, soil health and climate resilience?

RQ5	Can an explainable AI model communicate its recommendations effectively to farmers?

Research hypotheses

- H1: Historical environmental patterns can help identify periods of elevated agricultural risk.
- H2: Rotations incorporating suitable legumes or cover crops may offer greater soil-health benefits than continuous cultivation of the same vegetable crop.
- H3: Rotation strategies that account for local water availability will be more suitable than strategies based solely on crop preference.
- H4: Explainable recommendations will be more useful to farmers than recommendations without supporting evidence.

These hypotheses will be evaluated, not assumed to be true.

4. NASA datasets and research data sources

This is one of the most important parts of the project. We need to demonstrate that AGROINTEL AI uses genuine scientific observations.

The POWER of Earth Science Data | NASA Earthdata

4.1 NASA POWER

Phase 1 — Core dataset

- Temperature
- Solar radiation
- Wind speed
- Meteorological variables

Historical daily meteorological data are available from 1981 for supported parameters.

Official NASA POWER daily API

Docs+1

Soil Moisture Active Passive (SMAP) Project Assessment Report for Version 6 of the L4_SM Data Product

4.2 NASA SMAP

Phase 2

- Surface soil moisture
- Moisture anomalies
- Seasonal moisture changes

Used to investigate potential water stress.

SMAP data resources

NASA Satellite Captures Amazing 3D Videos of Rain, Snow | Live Science

4.3 NASA GPM

Phase 2

- Rainfall accumulation
- Heavy precipitation
- Seasonal rainfall distribution

Used to identify rainfall variability and potential waterlogging risks.

NASA GPM precipitation data directory

NDVI and EVI: Vegetation Indices (MODIS) | Climate Data Guide

4.4 NASA MODIS

Phase 3

- NDVI vegetation index
- Vegetation greenness
- Seasonal vegetation changes

Used as an indirect indicator of vegetation condition, not a direct measurement of crop health.

MODIS data resources

Camp Landsat - NASA Science

4.5 NASA Landsat

Phase 3

- Land-cover changes
- Vegetation conditions
- Field-level spatial observations

Useful for examining changes in agricultural land.

NASA Landsat programme

4.6 Supporting agricultural datasets

NASA observations alone are insufficient to determine which crop rotation is best.

We will also research:

Information	Potential source
Soil texture and properties	SoilGrids / ISRIC
Crop water requirements	FAO
Crop growing periods	FAO crop calendars
Vegetable characteristics	Agricultural research publications
Nitrogen fixation	Peer-reviewed agronomy studies
Crop prices	Local agricultural market information
Farmer constraints	Surveys and interviews

Useful references:

- ISRIC SoilGrids
- Food and Agriculture Organization
- FAO Crop Calendar

5. Vegetable crop-rotation research

We will use three strategies as the primary experimental scenarios.

Young tomatoes plantation furrows at Vegas Bajas del Guadiana, Spain

Strategy A — Conventional Vegetable Rotation

Example: Tomato → Cabbage → Lettuce.

Purpose:

- Establish a baseline.
- Examine nutrient demand.
- Identify potential pest and disease carryover.
- Compare against alternative rotations.

vegetable garden with tomatoes

Strategy B — Legume-Integrated Rotation

Example: Tomato → Beans → Cabbage.

Purpose:

- Introduce nitrogen-fixing legumes.
- Improve nitrogen cycling potential.
- Diversify crop families.
- Examine possible reductions in synthetic nitrogen requirements.

Important: Nitrogen benefits depend on biomass, nodulation, residue management and the amount of nitrogen actually retained in the soil.

The Plight of Clint and his Monoculture Practices | University of Maryland Extension

extension.umd.edu

Strategy C — Soil-Conservation Rotation

Example: Tomato → Cover Crop → Leafy Vegetables.

Purpose:

- Reduce bare-soil periods.
- Improve soil protection.
- Increase organic residue inputs.
- Potentially improve infiltration and reduce erosion.

The crop sequences are research examples, not universal recommendations. We will check crop-family relationships, local suitability, growing-season length and water requirements before assigning scores.

6. Research methodology

We will use a five-stage research methodology.

1. Stage 1 — Environmental data collection

Collect historical environmental observations for selected global locations.

Variables:

- Mean and maximum temperature.
- Precipitation.
- Soil moisture.
- Vegetation indices.
- Evapotranspiration where suitable data are available.

2. Stage 2 — Data processing and harmonisation
    - Standardise units.
    - Handle missing observations.
    - Aggregate daily measurements into seasonal indicators.
    - Align spatial and temporal resolutions.
    - Calculate historical anomalies.

All datasets will retain their original resolution and source metadata. We will not imply that resampling creates additional spatial detail.

3. Stage 3 — Environmental risk analysis

Calculate indicators such as:

- Temperature anomalies.
- Rainfall anomalies.
- Soil moisture anomalies.
- Vegetation condition anomalies.
- Consecutive dry periods.
- Extreme rainfall frequency.

These indicators help identify environmental conditions that may affect vegetable cultivation.

4. Stage 4 — Rotation modelling

Develop a multi-criteria decision model that compares three strategies using environmental conditions, crop characteristics and farmer priorities.

The model will explain why a strategy receives its score.

5. Stage 5 — Validation and evaluation
    - Compare model outputs with agricultural literature.
    - Conduct sensitivity testing.
    - Obtain feedback from agricultural experts or farmers.
    - Evaluate whether recommendations are understandable.
    - Document uncertainty and missing information.

7. AI and intelligent modelling approach

I recommend using a hybrid approach.

Rather than relying on a black-box AI model, AGROINTEL AI will combine statistical analysis, agricultural rules and multi-criteria decision-making.

Model components

Model	**Purpose**
Statistical analysis	Identify historical climate patterns
Anomaly detection	Identify unusual environmental conditions
Weighted decision model	Rank rotation strategies
Sensitivity analysis	Test how recommendations change when priorities change
Explainable AI layer	Explain the reasons behind recommendations

Proposed rotation scoring model

Each strategy receives a score from 0–100.

Where:

- \(S\) = Overall strategy score.
- \(w_i\) = Importance assigned to each criterion.
- \(r_i\) = Normalised score for that criterion.
- \(n\) = Number of evaluation criteria.

Initial research criteria:

Evaluation criterion	**Initial weight**
Soil-health potential	25%
Water-use suitability	20%
Climate resilience	20%
Pest and disease interruption	15%
Erosion reduction	10%
Economic suitability	10%
Total	100%

These are proposed starting weights, not scientifically established universal values. Farmer priorities will allow the weights to change.

8. Experimental design

To test the system, we should select three contrasting agricultural environments.

Aerial View of Agricultural Rows in Kern County, California Central Valley

Location 1 — California, USA

Research focus:

- Water availability.
- Irrigation dependence.
- Heat exposure.
- Drought resilience.

Reschedule work amid heat: IMD to Punjab farmers - The Tribune

Location 2 — Punjab, India

Research focus:

- Seasonal rainfall variability.
- Temperature stress.
- Crop-season transitions.
- Water management.

Farming at mountain

Location 3 — Cameron Highlands, Malaysia

Research focus:

- Tropical highland vegetable production.
- Rainfall patterns.
- Soil erosion.
- Excessive moisture.

These are proposed demonstration locations. We must verify actual agricultural fields, crop types and local growing calendars before finalising the test sites.

The same three rotation strategies will be evaluated at each location, with locally appropriate crops substituted where necessary.

This gives us at least nine scenario evaluations.

9. System evaluation and success criteria

We need to prove that AGROINTEL AI does more than display attractive graphs.

Evaluation area	**Proposed success measure**
Data reliability	NASA source, units and observation dates are traceable
Data processing	Missing values and anomalies are handled transparently
Rotation comparison	Three strategies evaluated at each test location
Explainability	Every recommendation includes supporting reasons

Sensitivity	Score changes correctly when farmer priorities change
Usability	Farmers can understand the main recommendations
Scientific validity	Agricultural claims reviewed against credible research
Transparency	Uncertainty and data gaps are clearly displayed

We will not claim improved crop yields or actual soil restoration unless supported by field trials or suitable validated studies.

10. Research timeline

PROPOSED 4-WEEK HACKATHON RESEARCH SCHEDULE

1. Week 1 — Literature review and dataset discovery
    - Research vegetable crop rotation.
    - Identify NASA datasets.
    - Select demonstration locations.
    - Establish scientific variables.
    - Prepare the research database.
2. Week 2 — Data collection and environmental analysis
    - Integrate NASA POWER.
    - Investigate SMAP and GPM.
    - Calculate environmental indicators.
    - Prepare charts and maps.
3. Week 3 — AI modelling and system development
    - Develop the scoring engine.
    - Implement three rotation strategies.
    - Add farmer priority controls.
    - Build the interactive dashboard.
4. Week 4 — Testing and research validation
    - Test multiple locations.

- Evaluate recommendations.
- Conduct sensitivity analysis.
- Prepare the final research report.
- Prepare the hackathon presentation.

11. Expected research outcomes

At the end of this project, we expect to produce:

1. A functioning global agricultural intelligence platform.
2. A collection of NASA-derived environmental indicators.
3. A historical climate and soil-moisture analysis.
4. An explainable vegetable crop-rotation recommendation model.
5. A comparison of three rotation strategies.
6. An interactive global demonstration map.
7. A scientific evaluation of model limitations and uncertainty.
8. A research report documenting methodology and findings.

12. Research limitations and risk management

It is important to acknowledge the following:

- Satellite soil-moisture observations have much coarser resolution than many individual vegetable fields.
- NASA POWER data are not direct measurements taken from a specific farm.
- NDVI may be affected by cloud cover, mixed vegetation and crop growth stages.
- Satellite observations cannot directly establish soil nitrogen, phosphorus or organic matter content.
- Crop rotation effects may take multiple growing seasons to become measurable.
- Economic recommendations require local price and market information.
- Crop disease and pest risks require crop-specific local evidence.

AGROINTEL AI will therefore provide decision support, not guaranteed crop outcomes.

13. Proposed final research statement

Research significance

AGROINTEL AI aims to demonstrate how NASA Earth observations and agricultural intelligence can be integrated into a practical, explainable crop-rotation decision-support platform. By analysing environmental conditions and comparing alternative vegetable crop rotations, the project seeks to help farmers identify opportunities for improving soil-health potential, managing water resources and strengthening resilience to climate variability.

Its primary contribution is the integration of satellite-derived environmental evidence, agricultural knowledge and farmer-defined priorities into a single accessible decision-making system.

14. Reference and literature collection

Start the formal research with these sources:

1. NASA POWER Daily API documentation — historical meteorological data.
2. NASA POWER Temporal APIs — daily, monthly and climatology services.
3. NASA GPM data directory — precipitation observations.
4. NASA SMAP mission — soil moisture observations.
5. NASA MODIS — vegetation observations.
6. ISRIC SoilGrids — soil property information.
7. FAO — agricultural and crop-management resources.

15. Immediate next research milestone

Before developing the AI model, we should create a proper scientific literature review and dataset specification.

I recommend starting with these three research areas:

- Research A: How climate variability affects vegetable production.
- Research B: How crop rotation, legumes and cover crops influence soil health.
- Research C: How NASA Earth observations can support agricultural decision-making.

The most important thing is to establish evidence for every scientific assumption that the recommendation engine will use.